Bangladesh University of Engineering and Technology

Department of Electrical and Electronic Engineering

EEE 414

Assignment

Lightning Protection System Design

Group 1

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Lightning Protection System Design

Project Overview

Building Type: 18-storey residential building

Unit conversion: $1\ ft = 0.3048\ m$

Roof Dimensions: $45\ ft\ \times 30\ ft = 13.7\ m\ \times 9.1$

Total Height = $18\ storey \times 10ft = 180\ ft = 54.864\ m$

Code Followed: BNBC 2020

Considering Location: Dhaka, Bangladesh (For Risk Index Assessment)

Risk Index Assessment

Height Calculation:

$18\ storey \times 10ft = 180\ ft = 54.864\ m$

Roof Area Calculation:

$45\ ft\ \times\ 30\ ft\ =\ 1350\ ft^2\ =\ 125.4\ m^2$

Index Breakdown:

A (Use of Structure) = 7 (Residential high-rise)

B (Construction Type) = 2 (Reinforced concrete)

C (Contents) = 2 (Domestic usage)

D (Isolation) = 2 (Urban dense area)

E (Terrain) = 2 (Flat terrain)

F (Height Factor) = 54.864 m (Height) > 53 m

G (Lightning Prevalence) = 21 (~80 thunderstorm days/year)

Total Risk Index = 7 + 2 + 2 + 2 + 2 + 30 + 21 = 66 > 40

Conclusion: Protection is mandatory due to height and high-risk index. So, **protection is essential.max**

Air Terminal Calculation

Using Rolling Sphere Method:

Assuming, Air Terminal Height, $h = 2\ m$

Assuming, protection radius, $r = 46\ ft \approx 14\ m$

Let, $d_{max} = Maximum\ Air\ Terminal\ Sepration$

So, $d_{max} = 2\sqrt{2rh - h^2}$

$$= 2\sqrt{2 \times 14 \times 2 - 2^2}\ m$$

$\therefore\ d_{max} = 14.42\ m$

So, maximum separation between air terminals should be 14.4 m.

So, number of air terminals along the length $= \lceil \frac{13.7}{14.4} \rceil = 1$

There are two sides along the length. So, along both sides there will be a total of 2 air terminals one on each side.

Similarly, along the width the number of air terminals will be $= \lceil \frac{9.1}{14.4} \rceil = 1$

There are two sides along the width. So, along both sides there will be a total of 2 air terminals one on each side.

So, total number of air terminals = 2+2 = 4

Final Design Decision: 4 Air Terminals placed at corners for complete coverage.

Down Conductor Calculation

Method 1 (Area Method):

For the first 80 m^2 there will be 1 down conductor and then for every extra 100 m^2 there will be 1 air terminal.

The area of the roof is $125.4\ m^2$.

So, total number of down conductors will be,

$$1 + \lceil \frac{125.4 - 80}{100} \rceil = 1 + 1 = 2$$

Method 2 (Perimeter Method):

1 air terminal is required for every 30 m of the perimeter.

$$Perimeter = 2(13.72 + 9.14) = 45.72\,m$$

Number of down conductors will be,

$$\left\lceil \frac{45.72}{30} \right\rceil = 2$$

Method 3 (Ideal Method):

Ideally, one down conductor per air terminal.

So, according to this, the number of down conductors will be 4 since we have 4 air terminals.

Final Decision: 4 down conductors selected (using the ideal method for best protection)

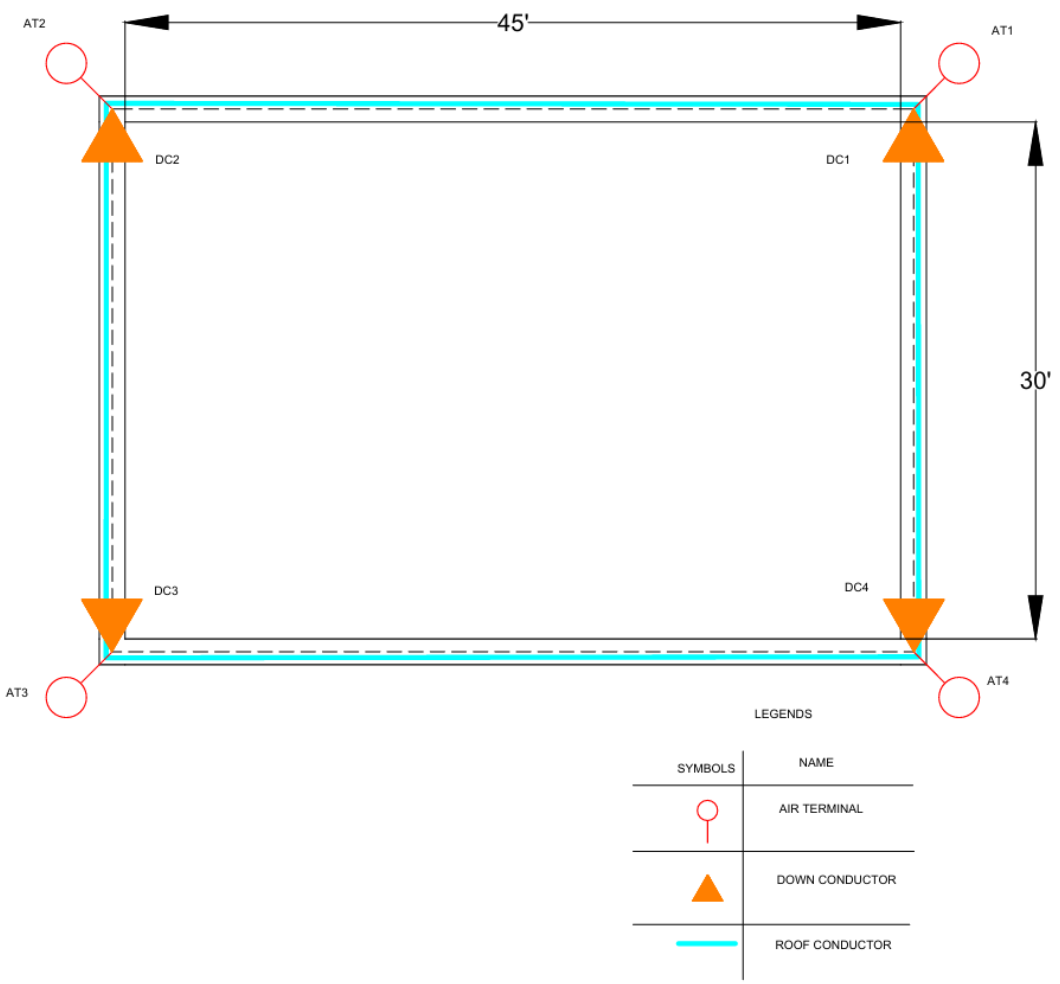

Figure: Roof Plan with Air Terminals and Down Conductors Connected through Roof Conductor

Protection Zone

Air Terminal height above roof = 2 m

Protection radius = 14 m (cone method)

For full roof coverage, roof conductors covers the entire 13.7 × 9.1 m roof.

According to the code, No part of roof should be more than 9 m from nearest horizontal conductor.

Max distance in our 13.7 × 9.1 m roof = $\frac{\text{Diagonal}}{2} = \frac{\sqrt{13.7^2+9.1^2}}{2} \approx$ 8.3 m (which is under 9 meters.)

So, roof is fully protected.

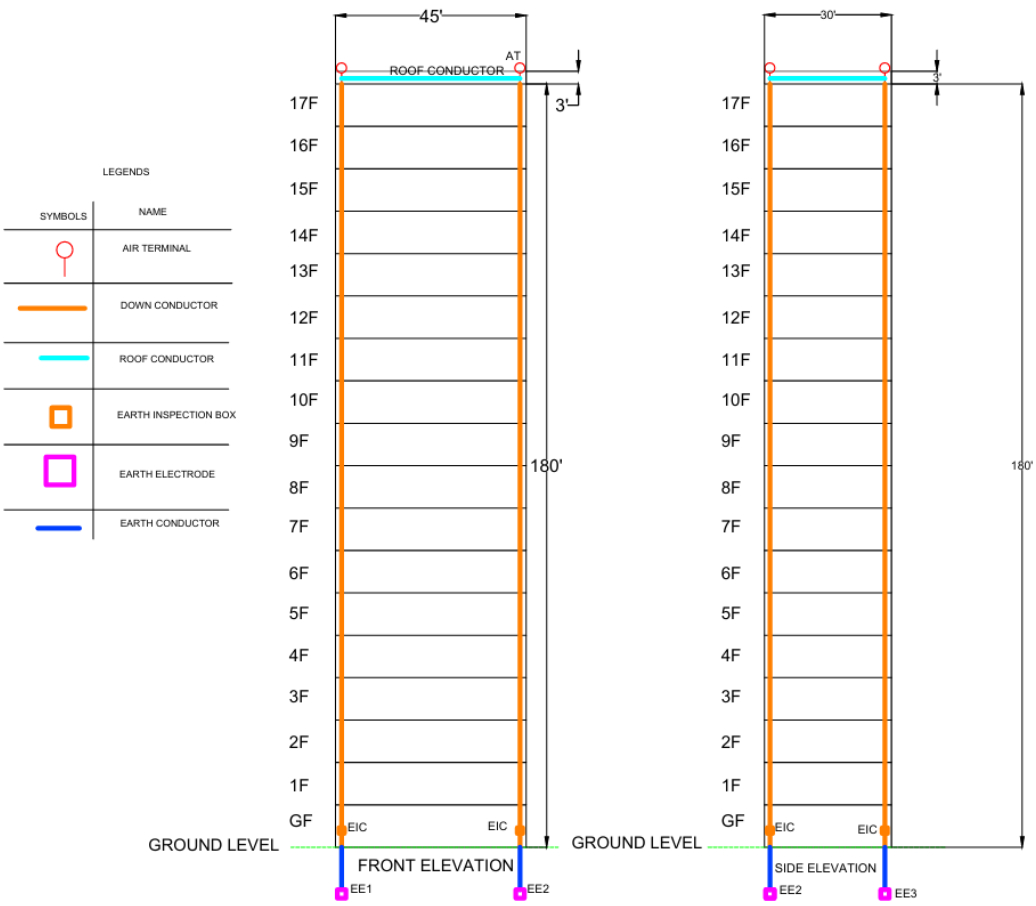

Figure: Front and Side Elevation Plan of the Down Conductors including Earth Electrode and Earth Inspection Box 1000 mm (1m) above Ground Level

Rolling Sphere Method

Sphere radius R = 46 ft = 14 m

Air terminal tip height = 2 m

Depth of Penetration, $h_p = r - \sqrt{r^2 - (\frac{d_{max}}{2})^2} = 14 - \sqrt{14^2 - (\frac{14.42}{2})^2} =$ 2 m (edge condition)

So, roof is protected but placing it over the parapet will give extra height for more protection and bonding is required.

Conclusion: Roof protected, bonding required for walls

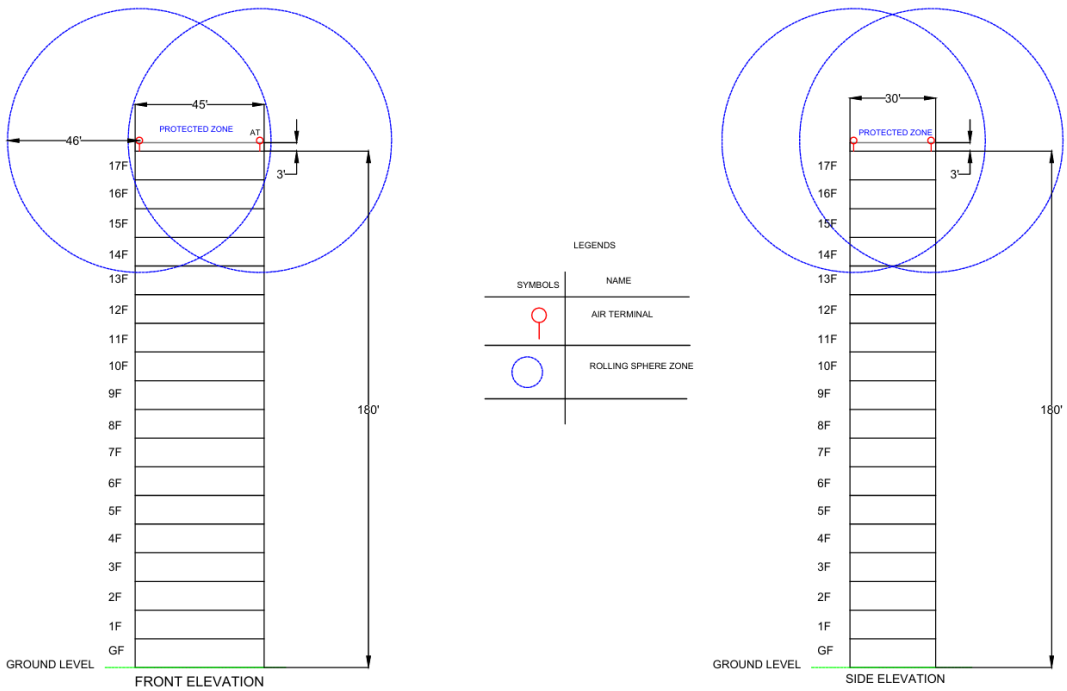

Figure: Protected Zones by Rolling Sphere Method

Clearance Calculation

Formula: $\boldsymbol{D = 0.3R + \frac{H}{15n}}$

Here, D = Clearance

R = Earth Resistance = 2Ω (for high rise above 10 storeys)

H = Height of the building

n = Number of Down Conductors

$$D = 0.3(2) + \frac{54}{15 \times 4}$$

D = 0.6 + 0.9 = 1.5 m

Conclusion: Bonding used instead of clearance

Earthing System

Earth resistance requirement: ≤2 Ω (For high rise above 10 storeys)

Number of electrodes: 4 (one per down conductor)

Type: GI pipe electrode (50 mm diameter)

Component Specifications

Air Terminal: 12 mm copper rod, 2 m height

Roof Conductor: 20 mm × 3 mm copper strip

Down Conductor: 20 mm × 3 mm copper strip

Earth Electrode: 50 mm GI pipe